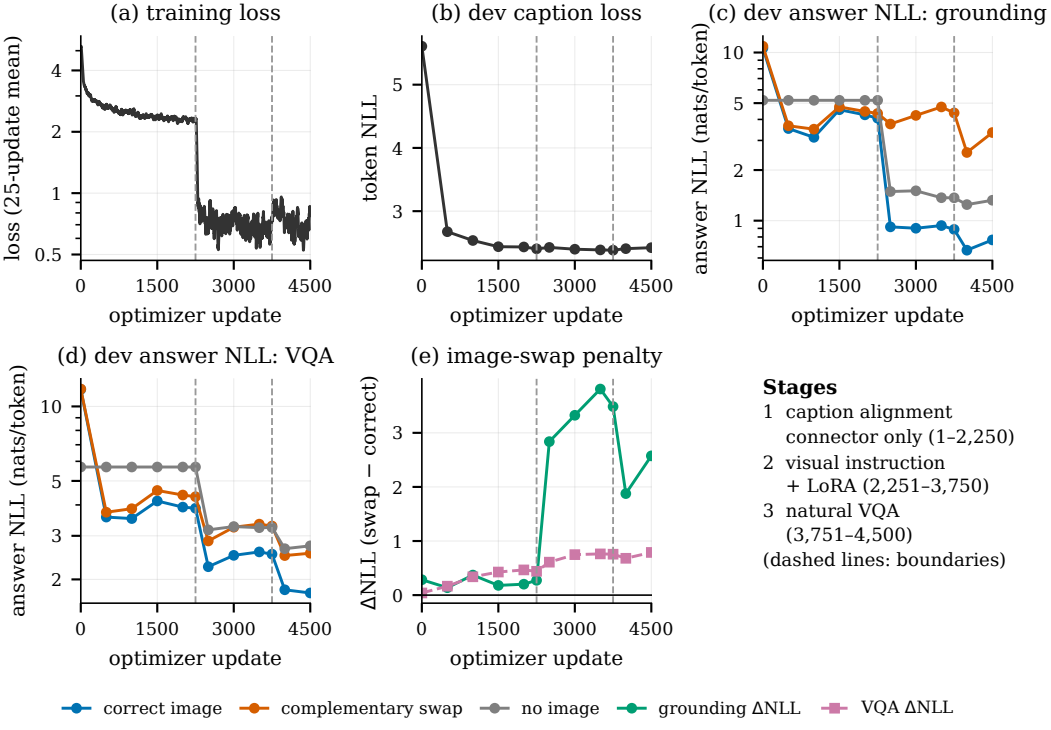
(a) training loss
loss (25-update mean)
optimizer update
(b) dev caption loss
token NLL
optimizer update
(c) dev answer NLL: grounding
answer NLL (nats/token)
optimizer update
(d) dev answer NLL: VQA
answer NLL (nats/token)
optimizer update
(e) image-swap penalty
ΔNLL (swap − correct)
optimizer update
Stages
1 caption alignment
connector only (1–2,250)
2 visual instruction
+ LoRA (2,251–3,750)
3 natural VQA
(3,751–4,500)
(dashed lines: boundaries)
correct image
complementary swap
no image
grounding ΔNLL
VQA ΔNLL